Auto Encoders Intuition

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Plan of Attack

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Plan of Attack

What we will learn in this section:

- Auto Encoders
- Training of an Auto Encoder
- Overcomplete Hidden Layers
- Sparse Autoencoders
- Denoising Autoencoders
- Contractive Autoencoders
- Stacked Autoencoders
- Deep Autoencoders

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Auto Encoders

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Auto Encoders

Supervised	Artificial Neural Networks	Used for Regression & Classification
	Convolutional Neural Networks	Used for Computer Vision
	Recurrent Neural Networks	Used for Time Series Analysis
Unsupervised	Self-Organizing Maps	Used for Feature Detection
	Deep Boltzmann Machines	Used for Recommendation Systems
	AutoEncoders	Used for Recommendation Systems

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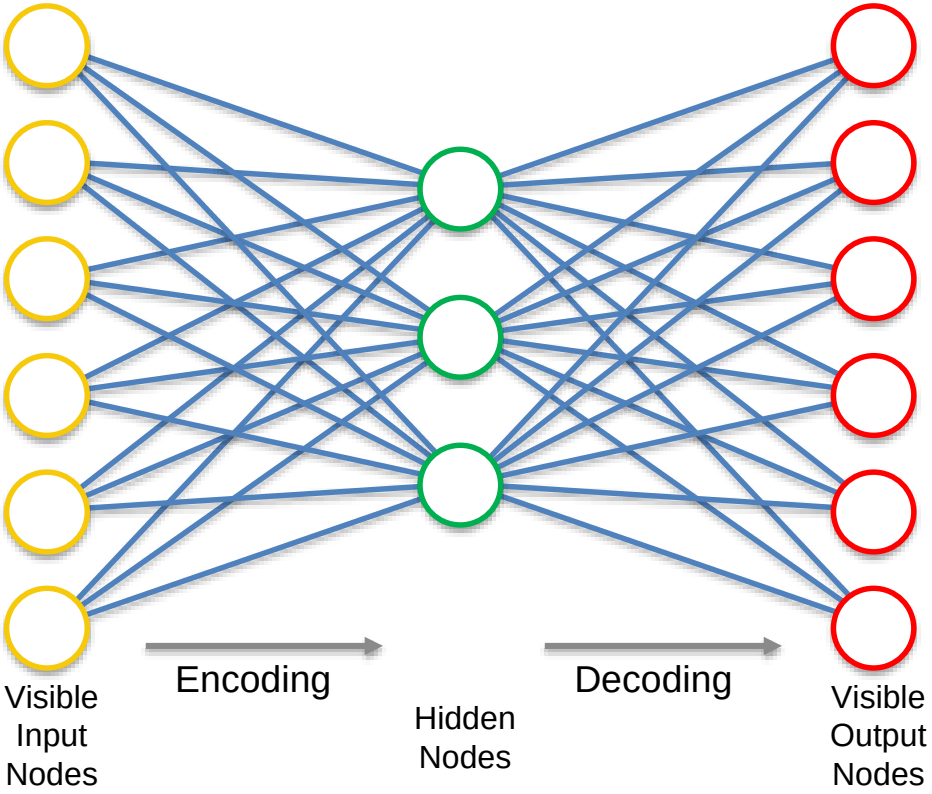
Auto Encoders



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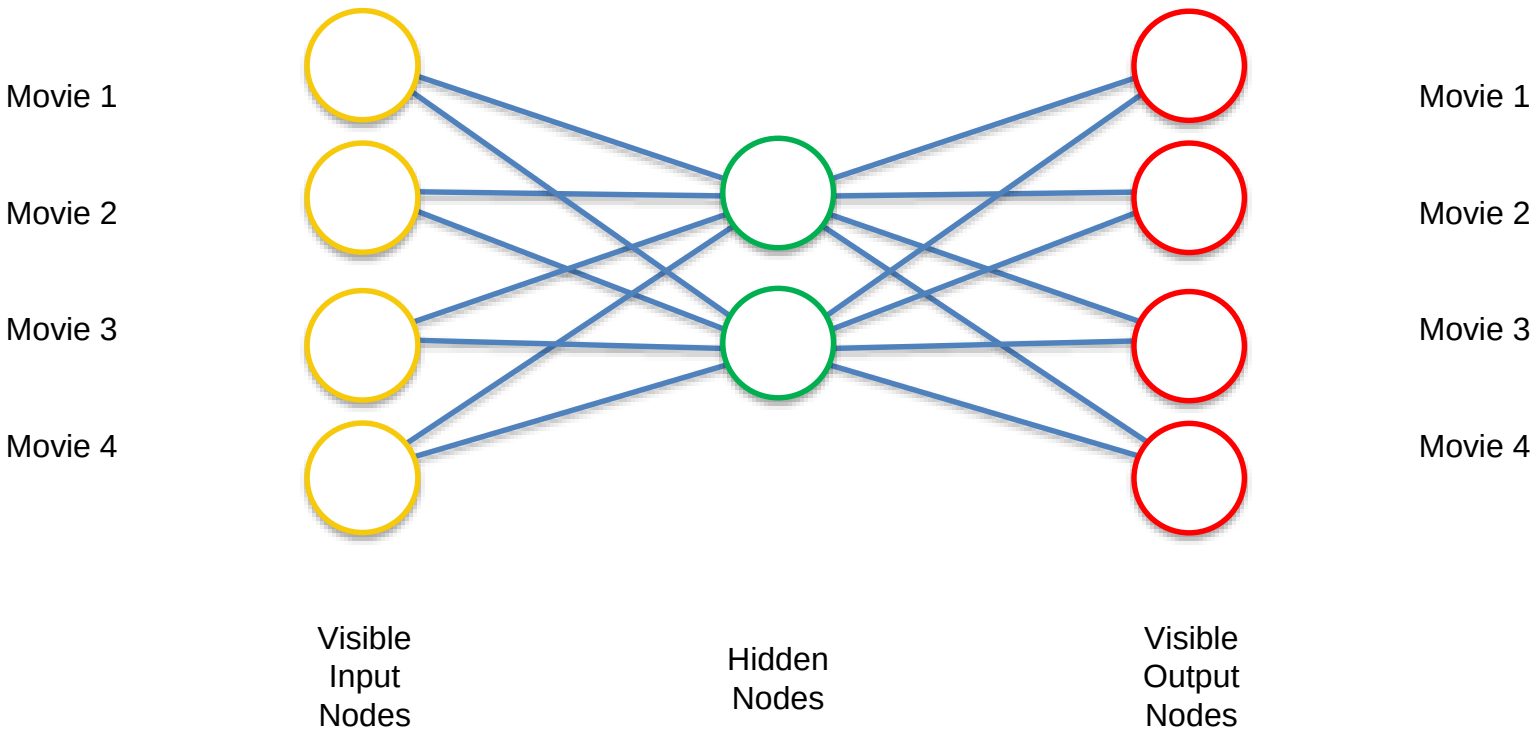
Auto Encoders



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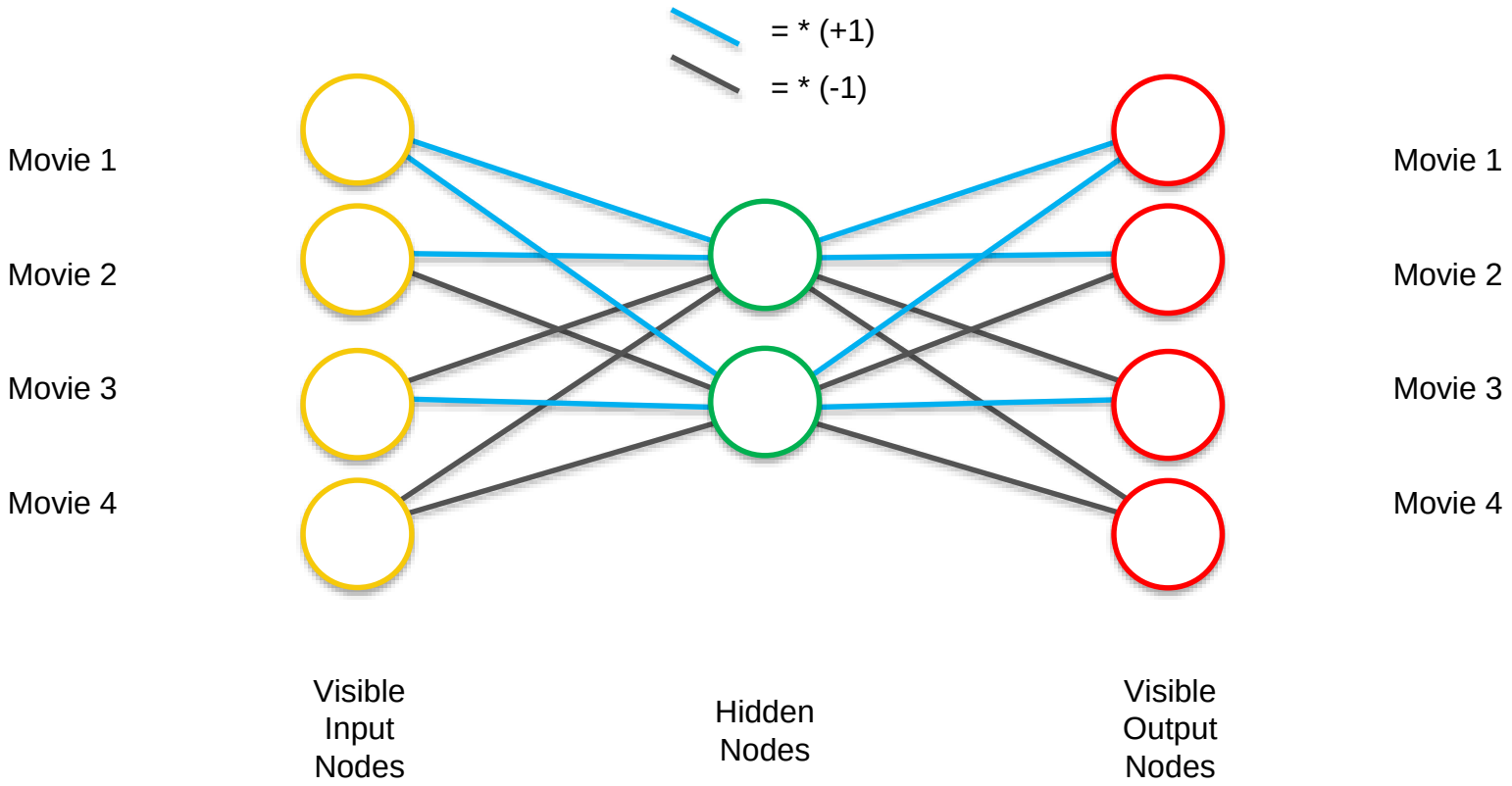

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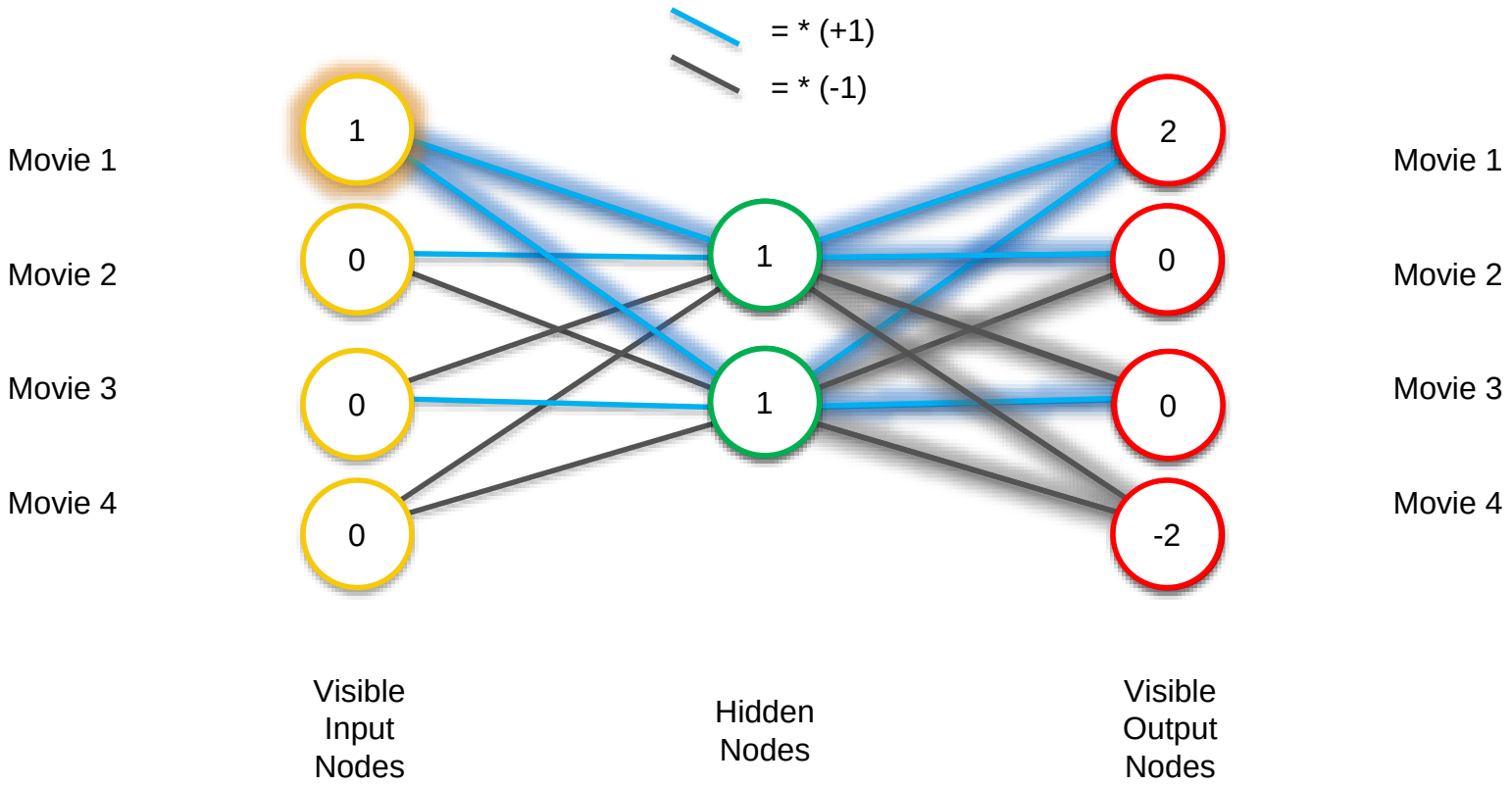
Auto Encoders



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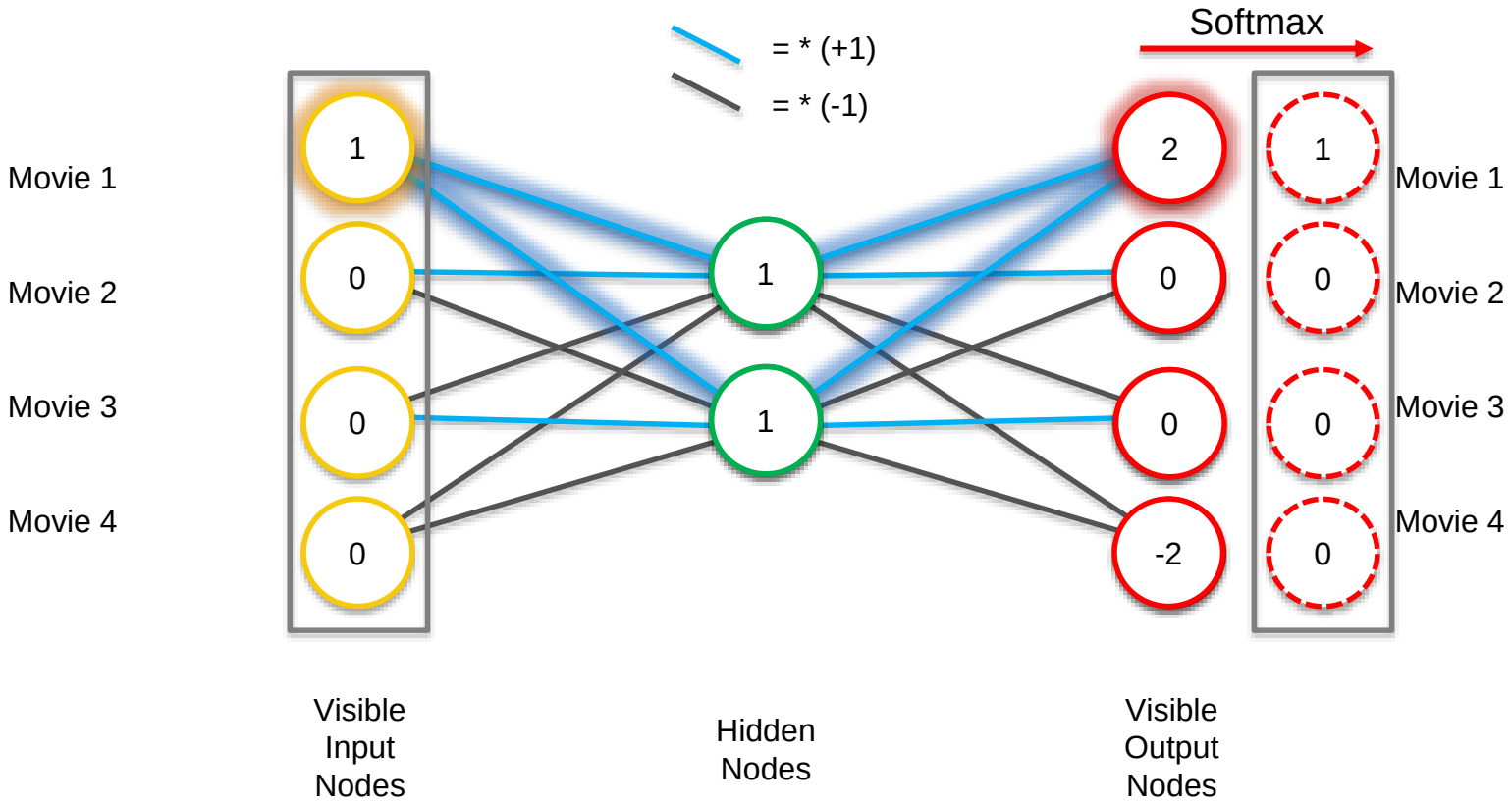
Auto Encoders



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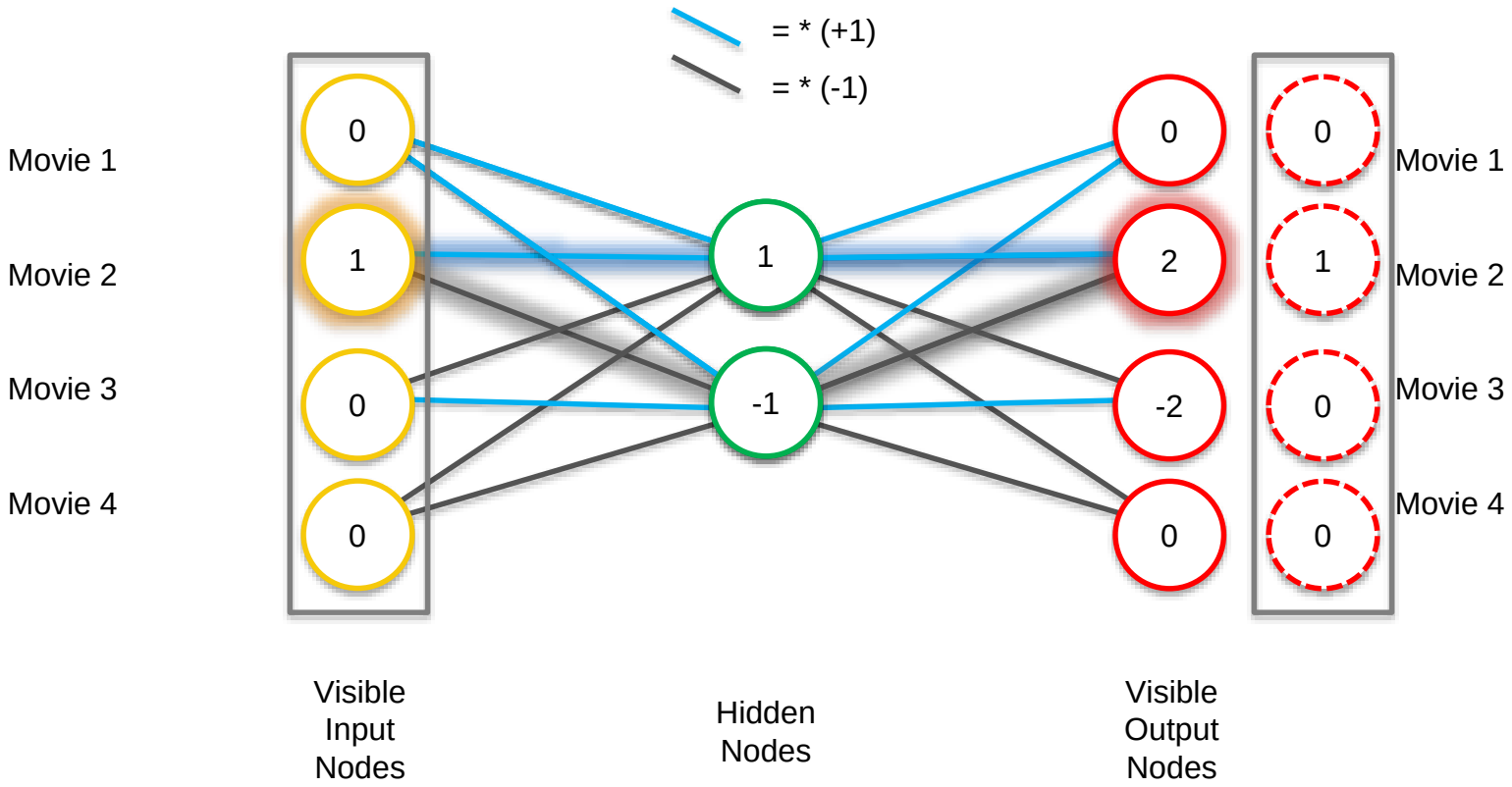
Auto Encoders



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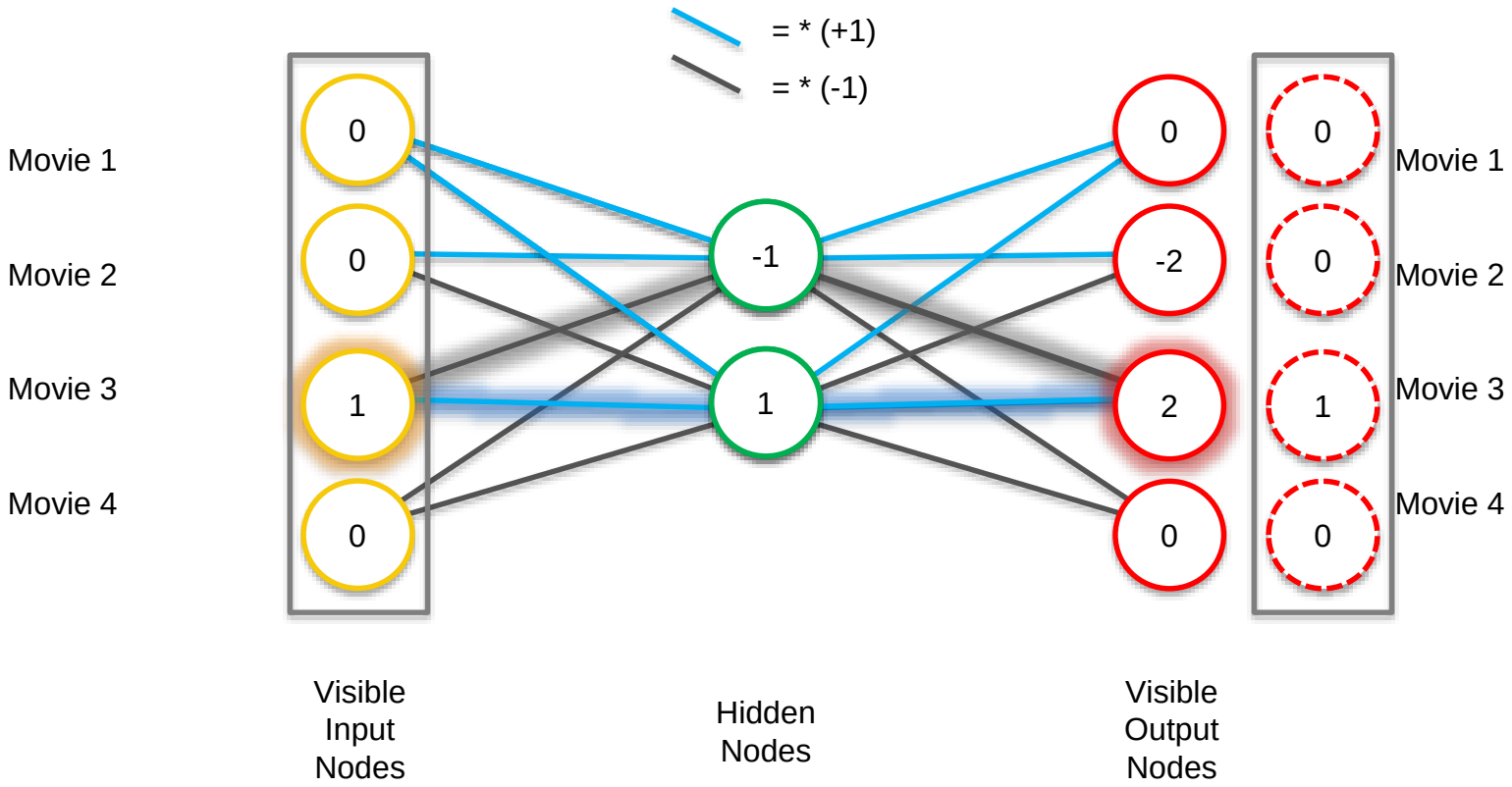
Auto Encoders



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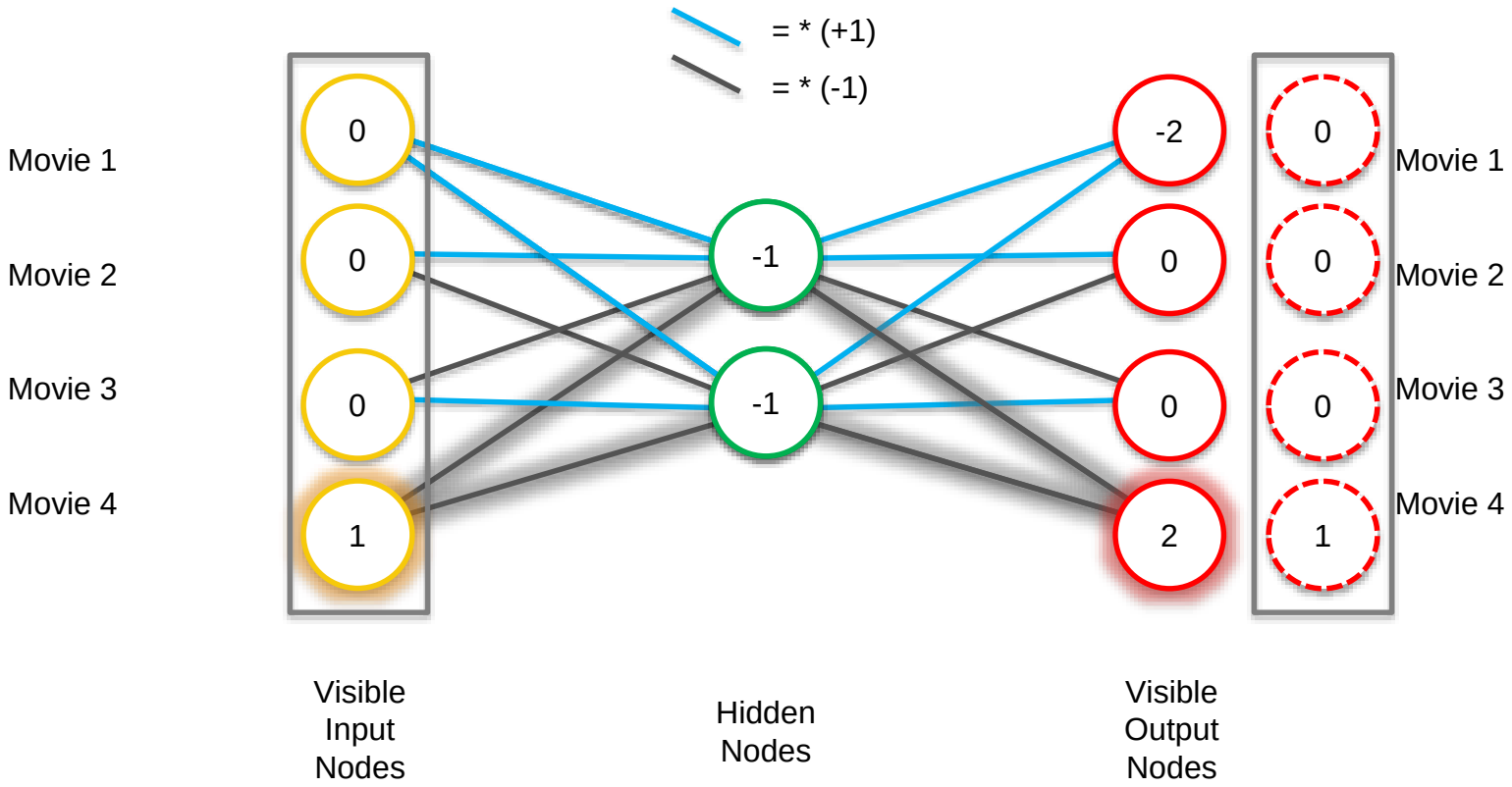
Auto Encoders



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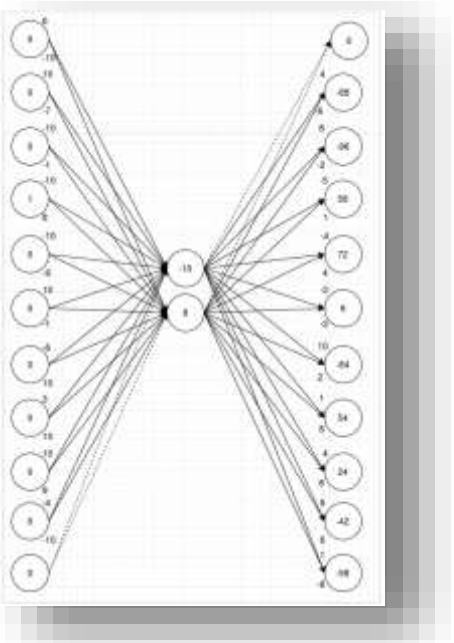
Auto Encoders

Additional Reading:

Neural Networks Are Impressively Good At Compression

By Malte Skarupke (2016)



Link:

https://probablydance.com/2016/04/30/neural-networks-are-impressively-good-at-compression/

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A Note on Biases

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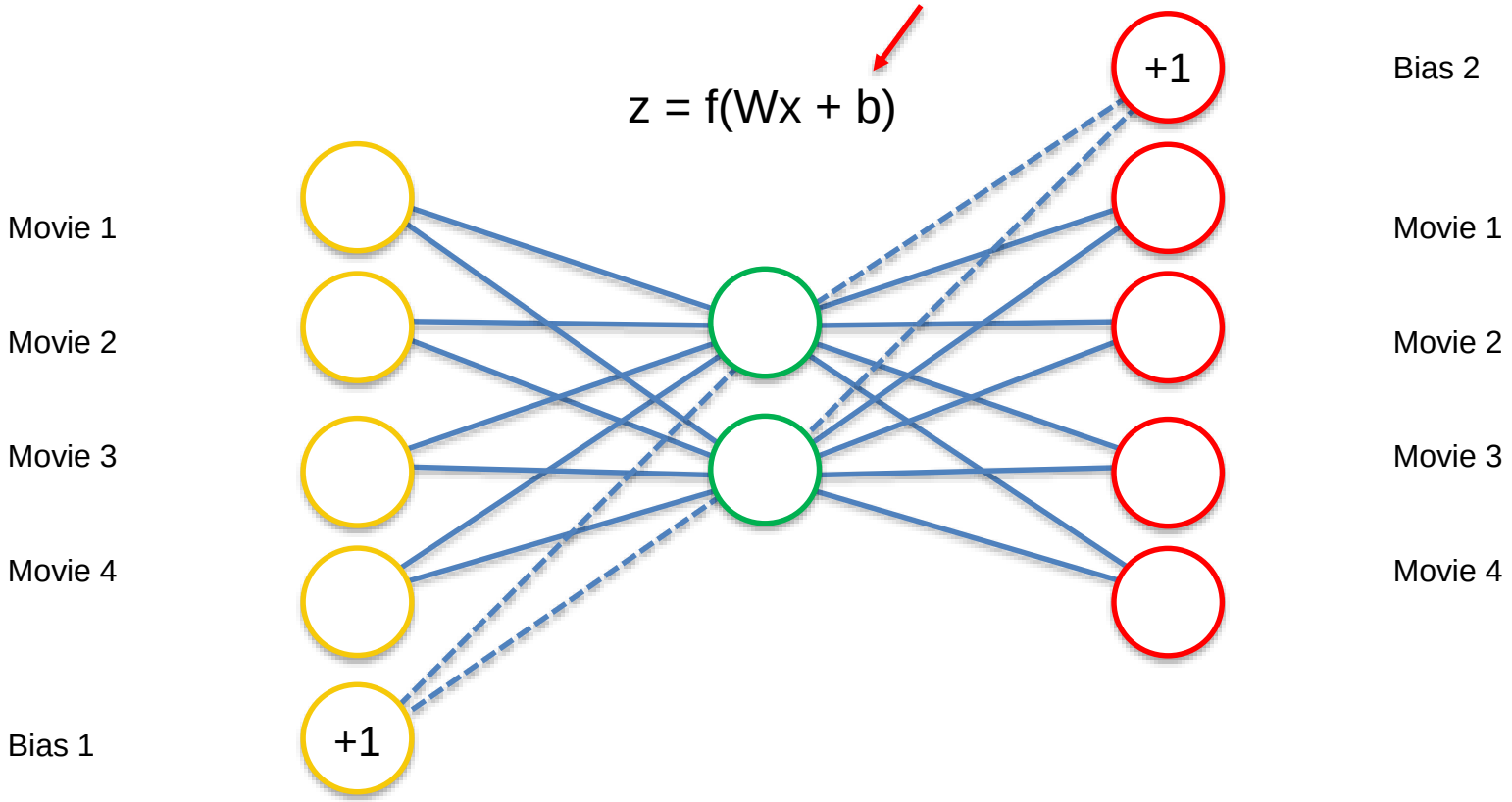
Auto Encoders



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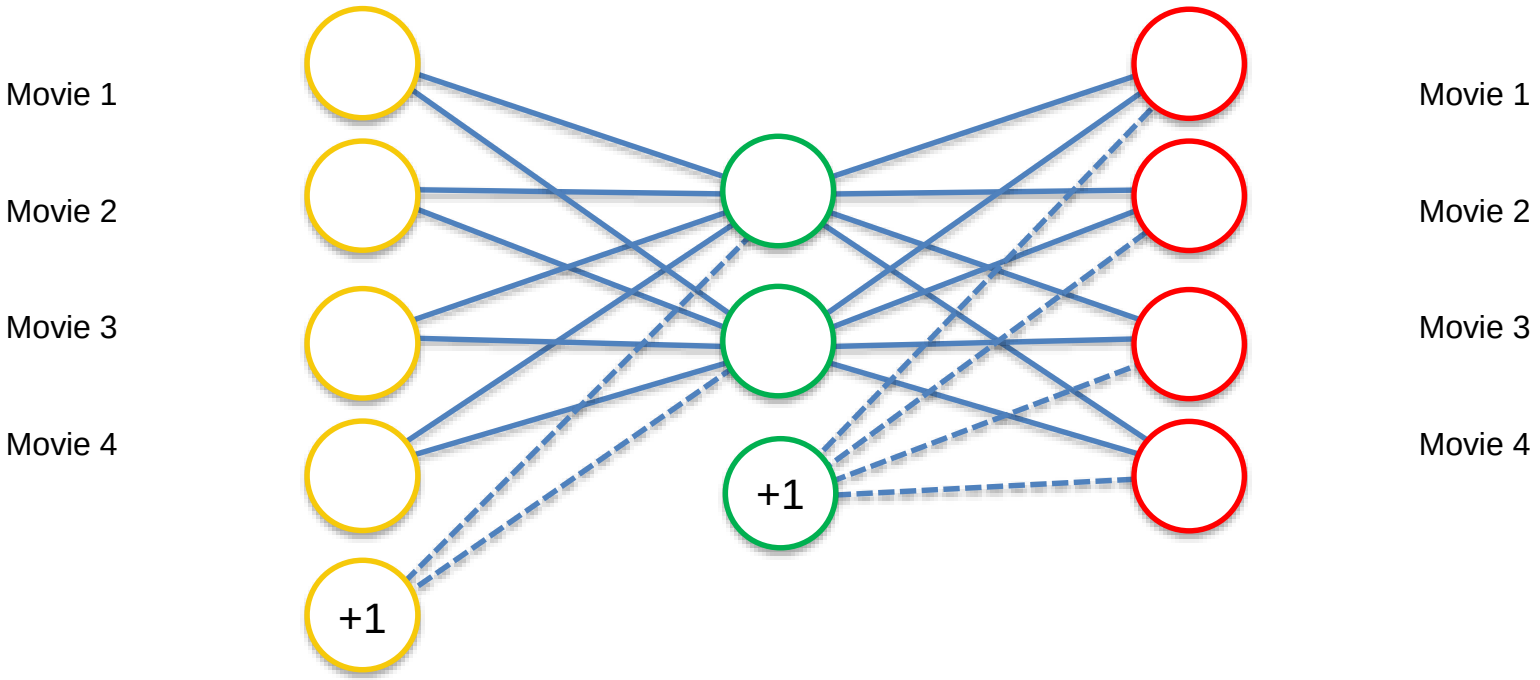
Auto Encoders

$$z = f(Wx + b)$$



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Training an Auto Encoder

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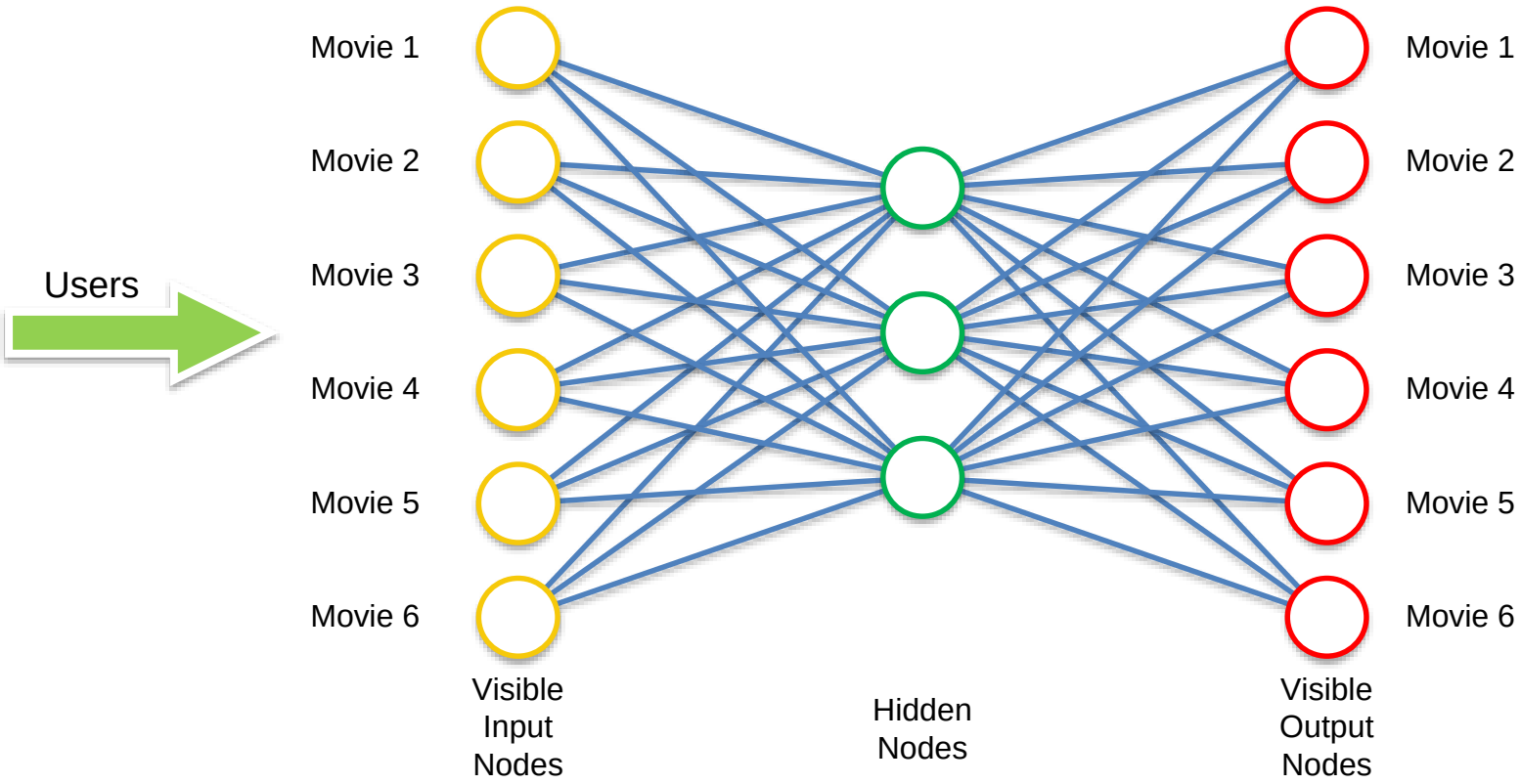
Training an Auto Encoder



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Training an Auto Encoder

STEP 1: We start with an array where the lines (the observations) correspond to the users and the columns (the features) correspond to the movies. Each cell (u, i) contains the rating (from 1 to 5, 0 if no rating) of the movie i by the user u.

STEP 2: The first user goes into the network. The input vector $x = (r_1, r_2, \ldots, r_m)$ contains all its ratings for all the movies.

STEP 3: The input vector x is encoded into a vector z of lower dimensions by a mapping function f (e.g: sigmoid function):

$z = f(Wx + b)$ where W is the vector of input weights and b the bias

STEP 4: z is then decoded into the output vector y of same dimensions as x, aiming to replicate the input vector x.

STEP 5: The reconstruction error $d(x, y) = ||x-y||$ is computed. The goal is to minimize it.

STEP 6: Back-Propagation: from right to left, the error is back-propagated. The weights are updated according to how much they are responsible for the error. The learning rate decides by how much we update the weights.

STEP 7: Repeat Steps 1 to 6 and update the weights after each observation (Reinforcement Learning). Or:
Repeat Steps 1 to 6 but update the weights only after a batch of observations (Batch Learning).

STEP 8: When the whole training set passed through the ANN, that makes an epoch. Redo more epochs.

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Training an Auto Encoder

STEP 1

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1

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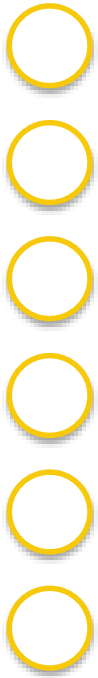
Training an Auto Encoder

STEP 2

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



Visible Input Nodes

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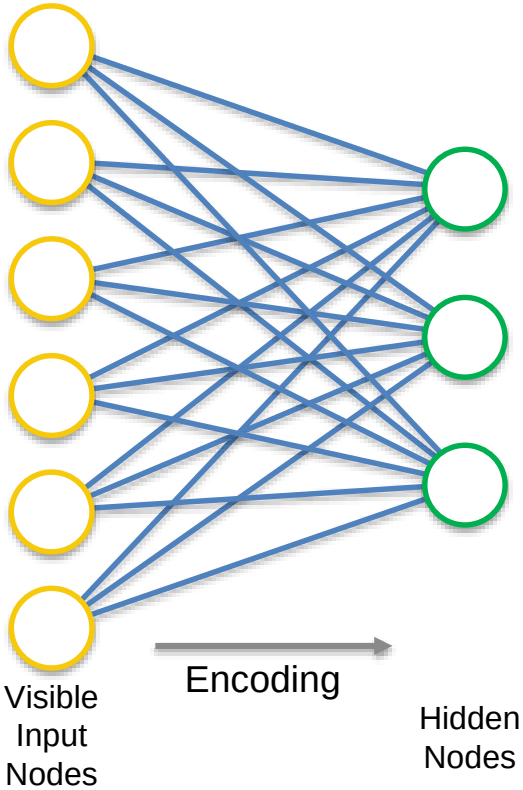
Training an Auto Encoder

STEP 3

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



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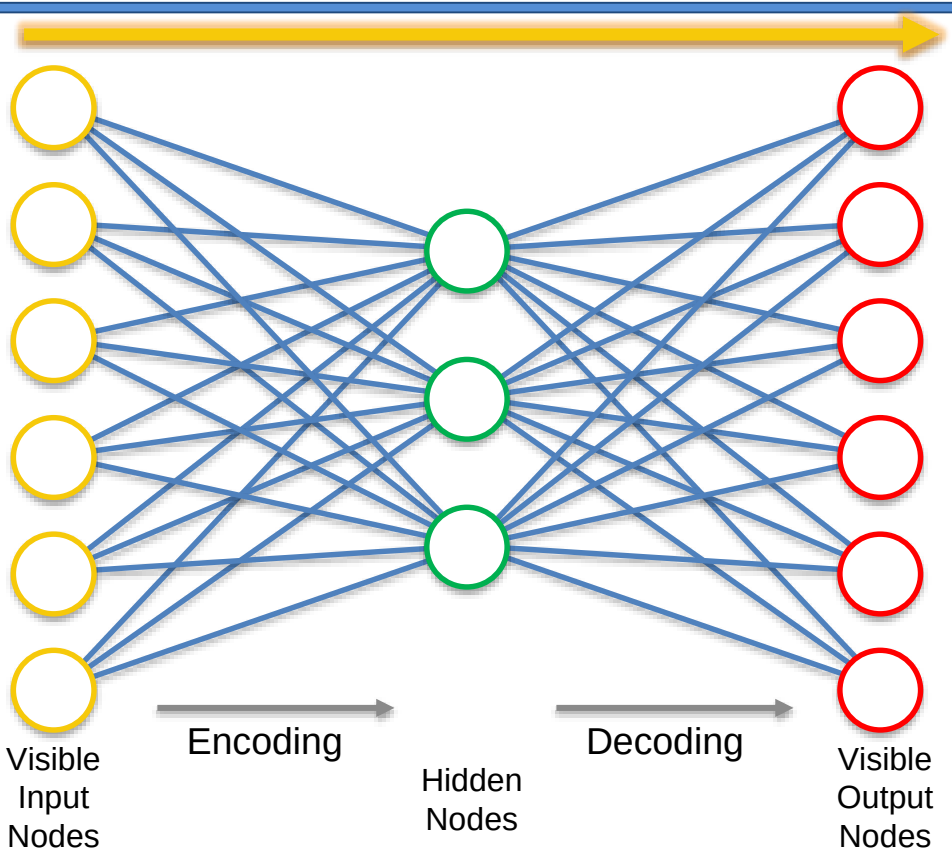
Training an Auto Encoder

STEP 4

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



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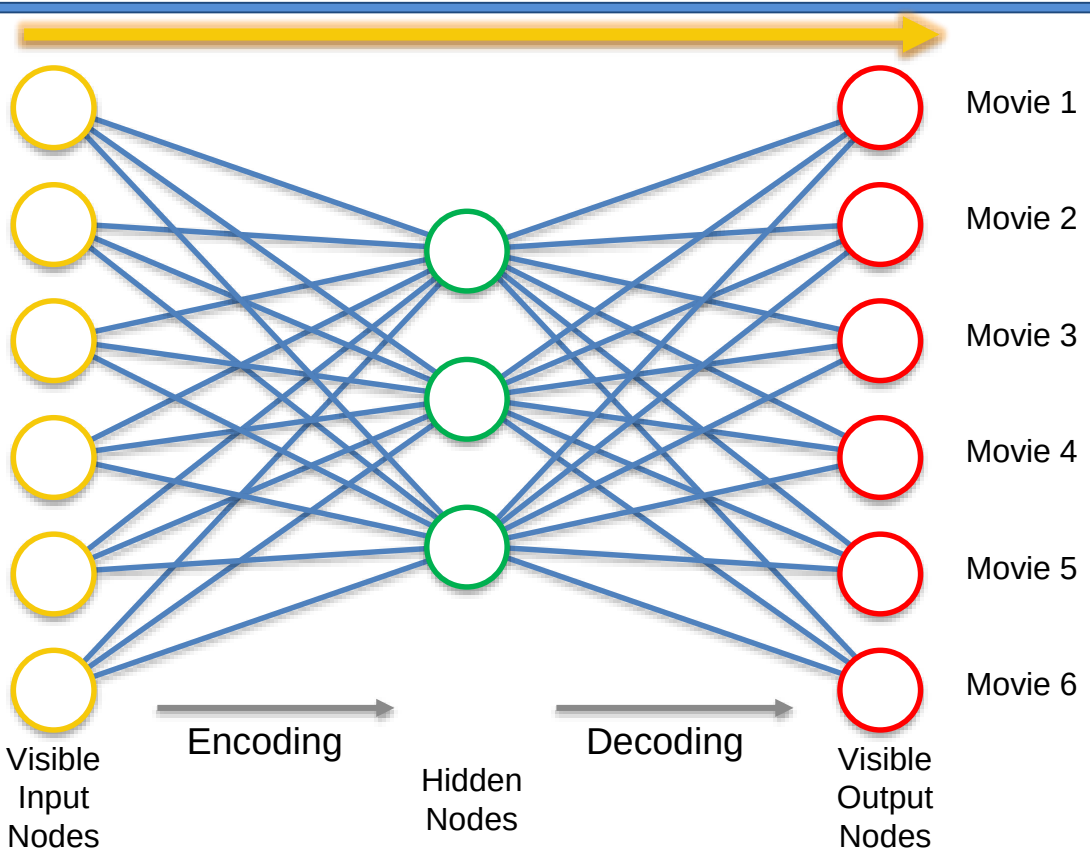
Training an Auto Encoder

STEP 5

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



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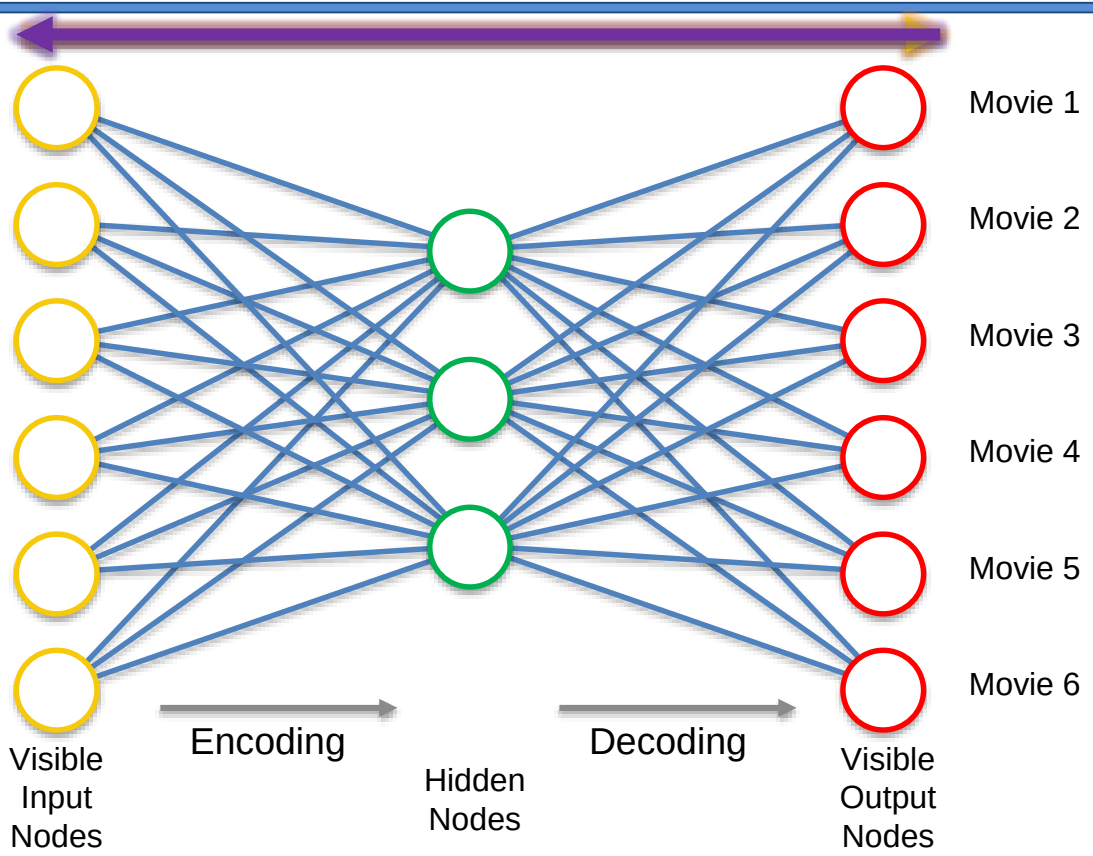
Training an Auto Encoder

STEP 6

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



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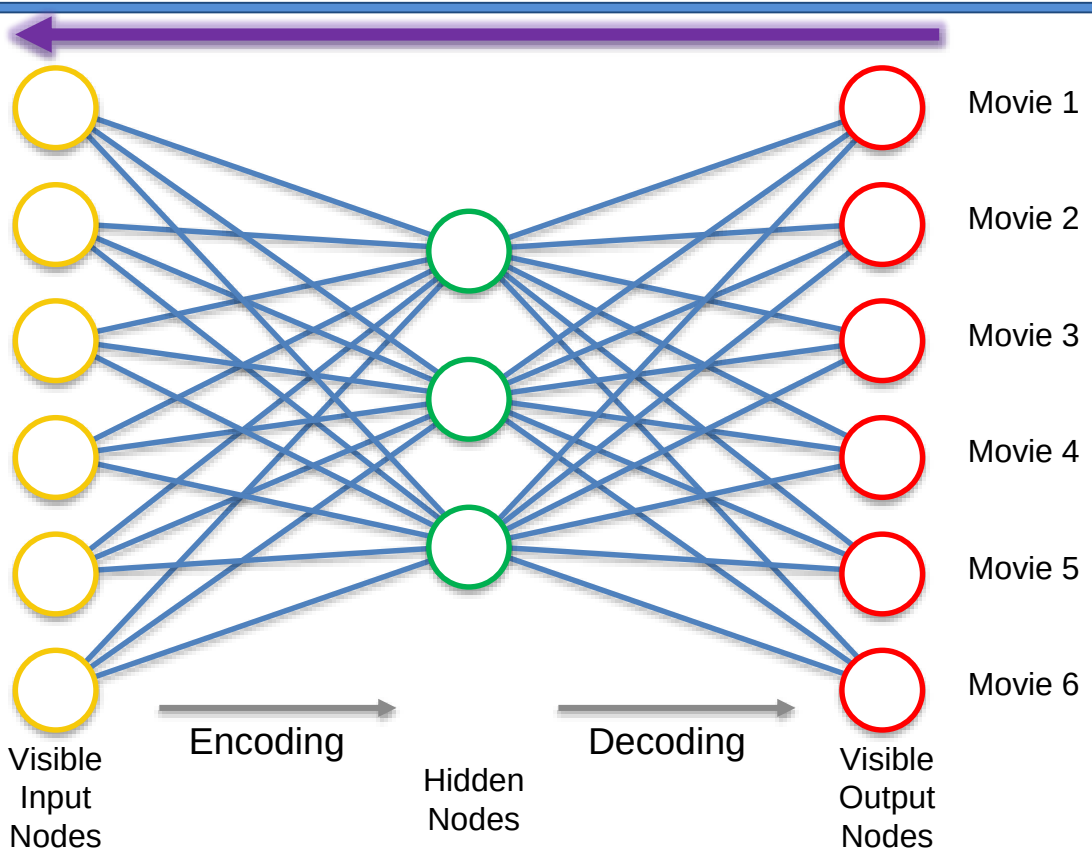
Training an Auto Encoder

STEP 7

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1



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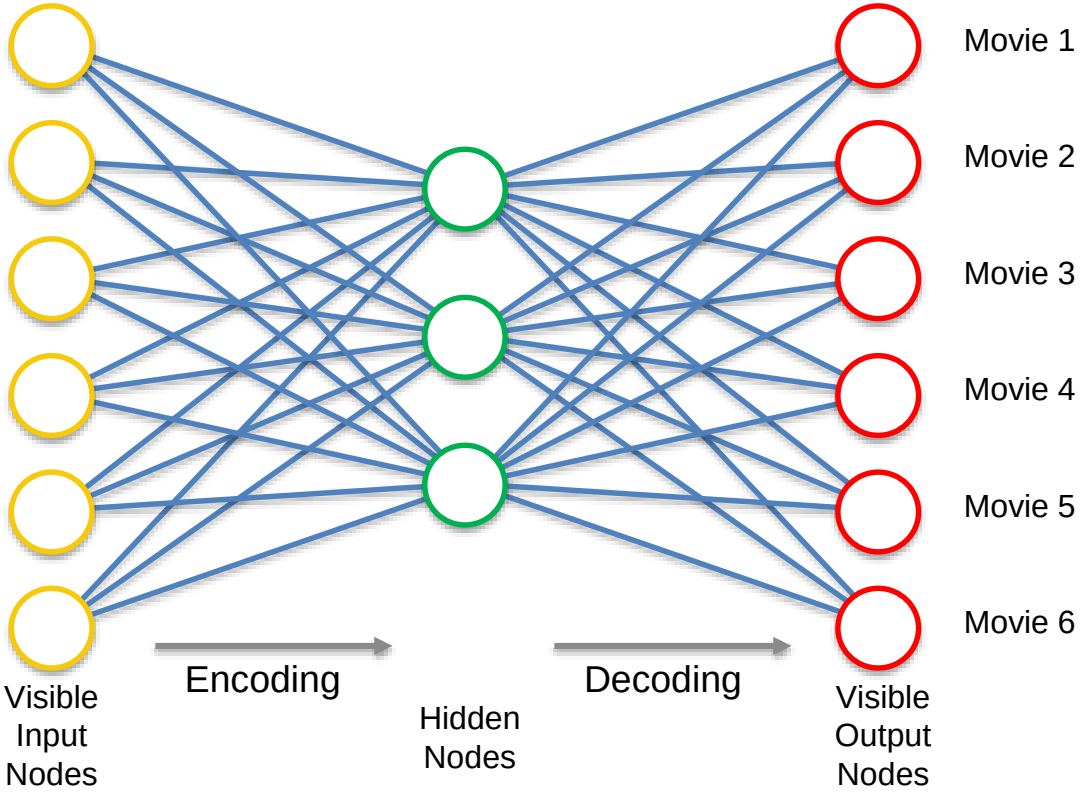
Training an Auto Encoder

STEP 8

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie 6
User 1	1	0		1	1	1
User 2	0	1	0	0	1	0
User 3		1	1	0	0	
User 4	1	0	1	1	0	1
User 5	0		1	1		1
User 6	0	0	0	0	1	
User 7	1	0	1	1	0	1
User 8	0	1	1		0	1
User 9		0	1	1	1	1
User 10	1		0	0		0
User 11	0	1	1	1	0	1

Repeat



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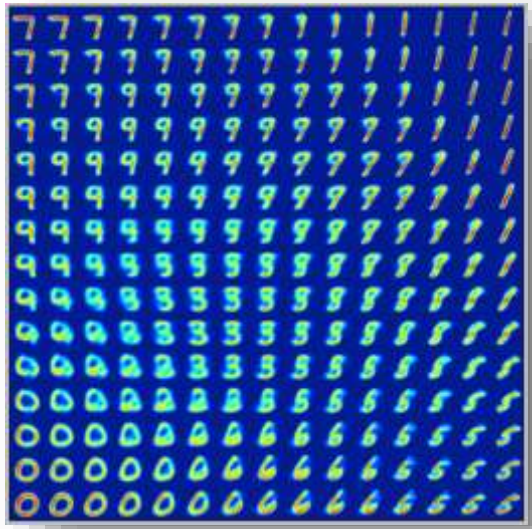
Training an Auto Encoder

Additional Reading:

Building Autoencoders in Keras

By Francois Chollet (2016)



Link:

https://blog.keras.io/building-autoencoders-in-keras.html

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Overcomplete Hidden Layers

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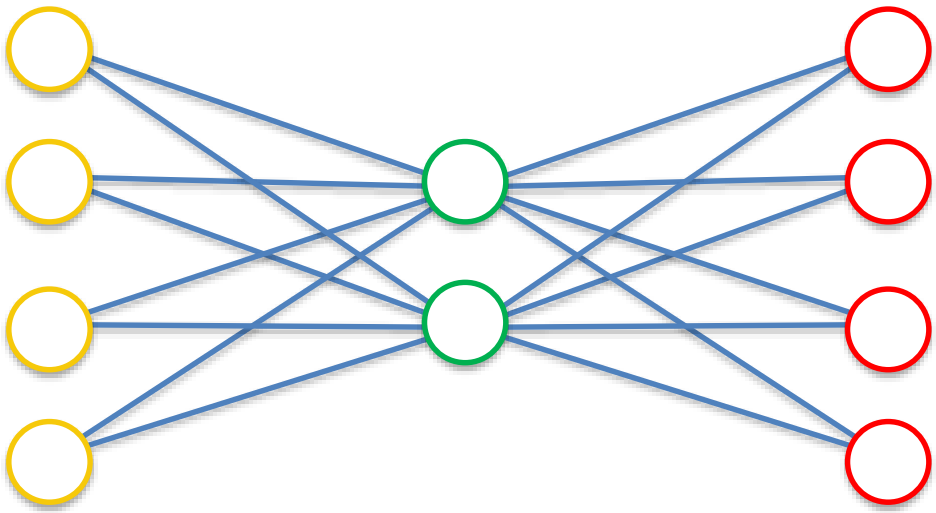
Overcomplete Hidden Layers



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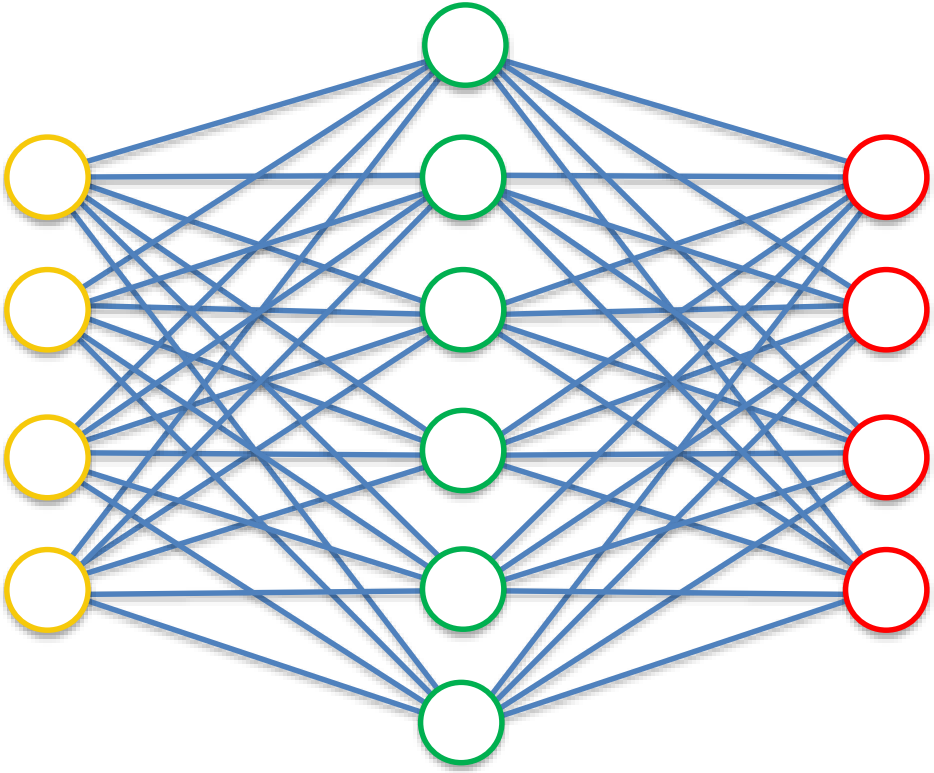
Overcomplete Hidden Layers



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Sparse Autoencoders

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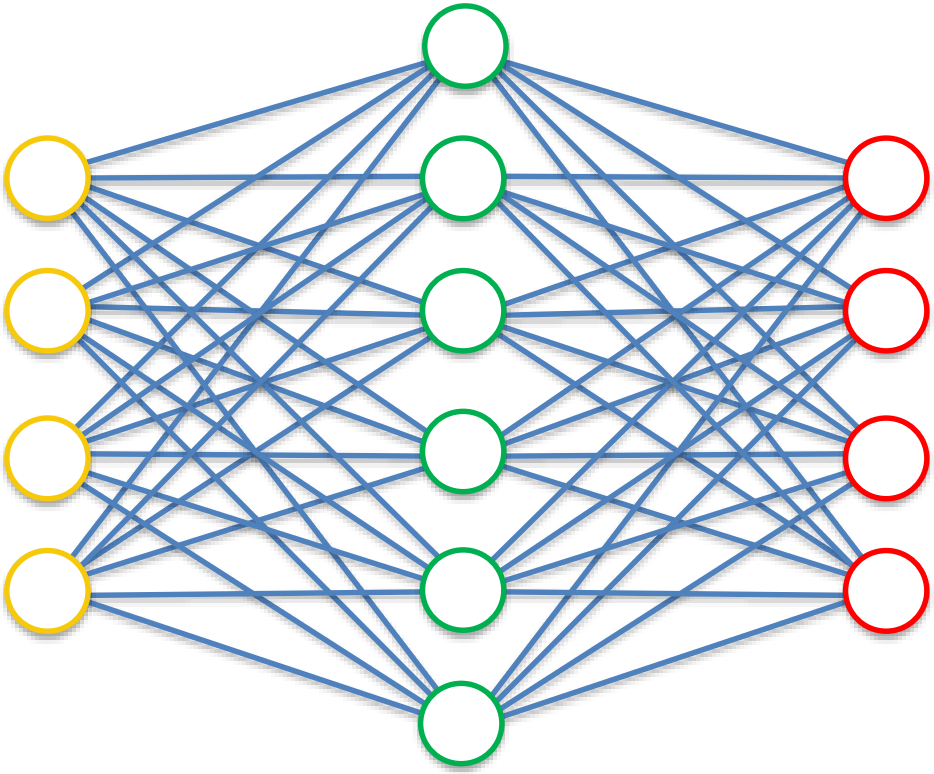
Sparse Autoencoders



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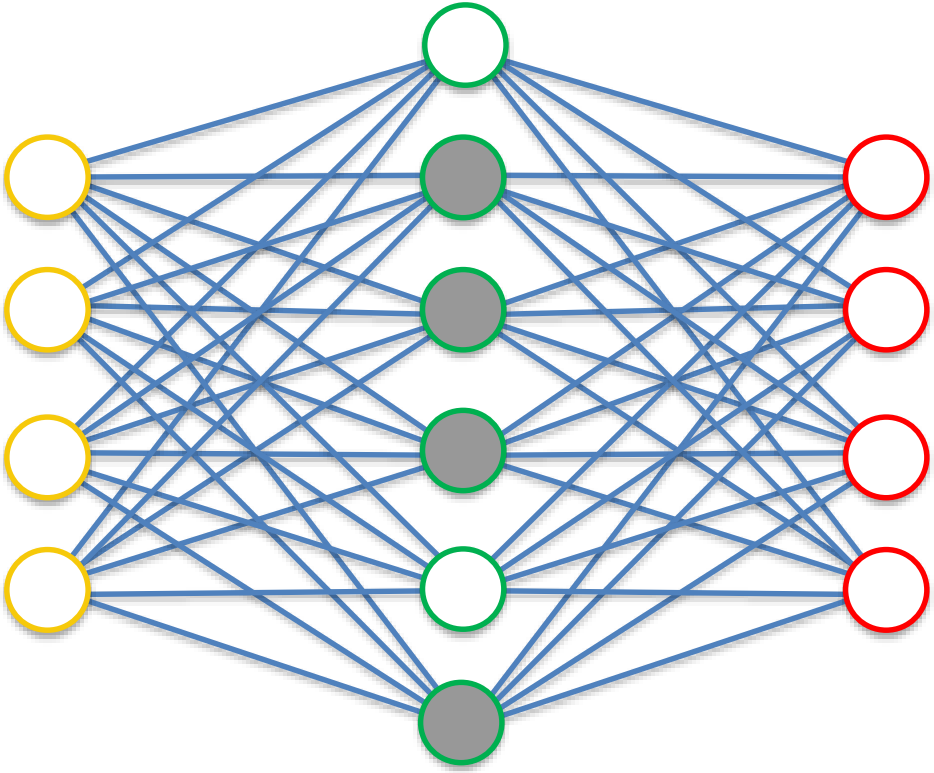

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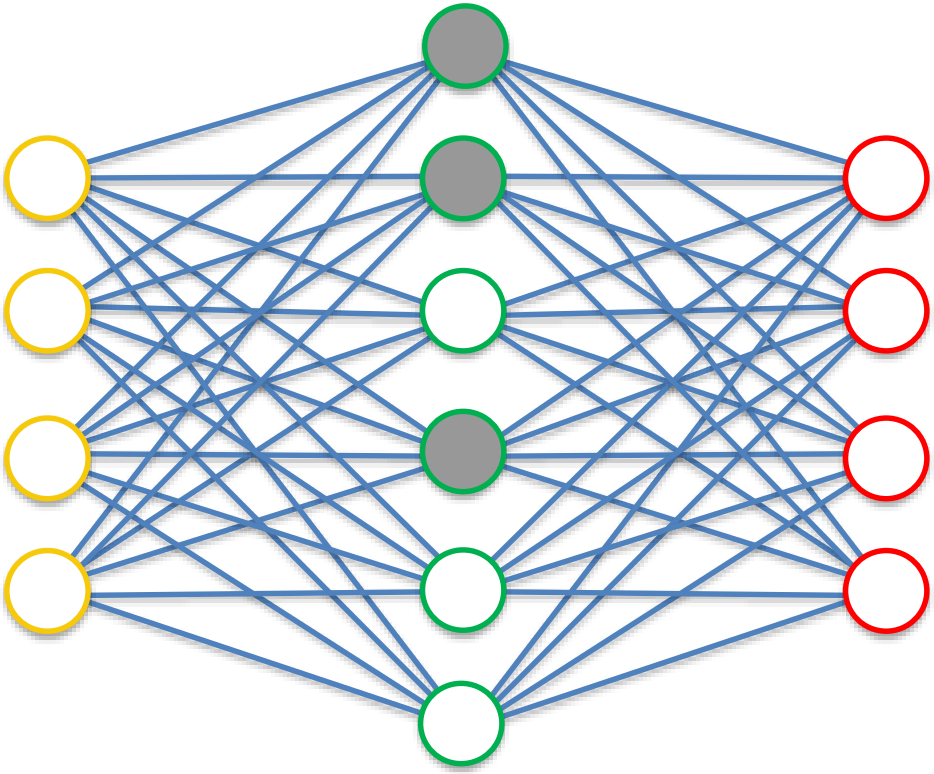

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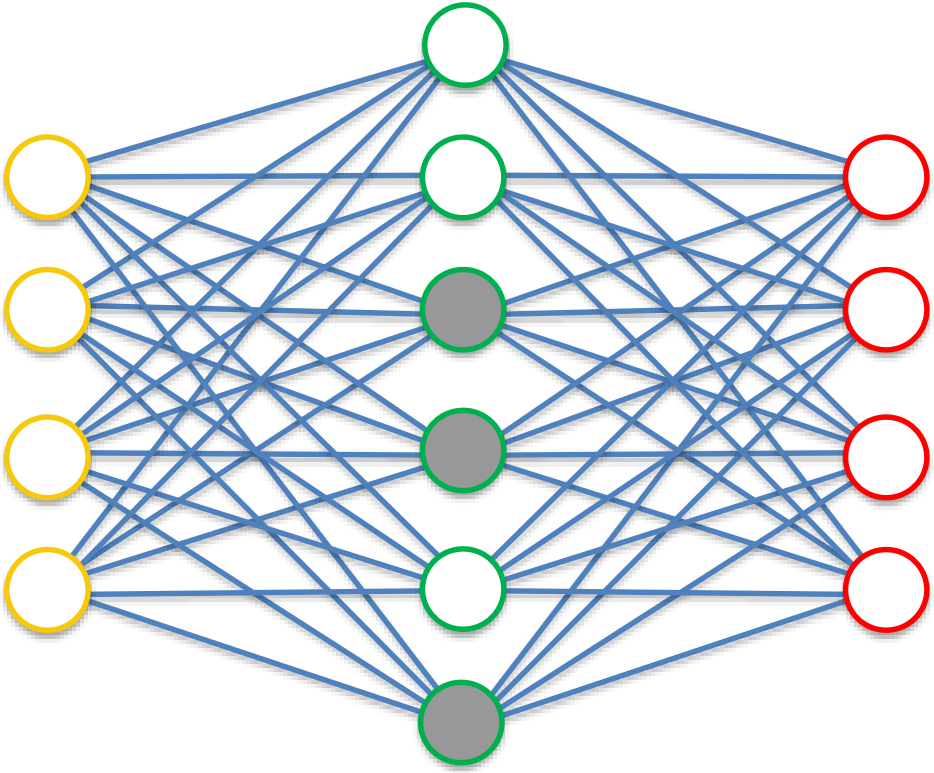

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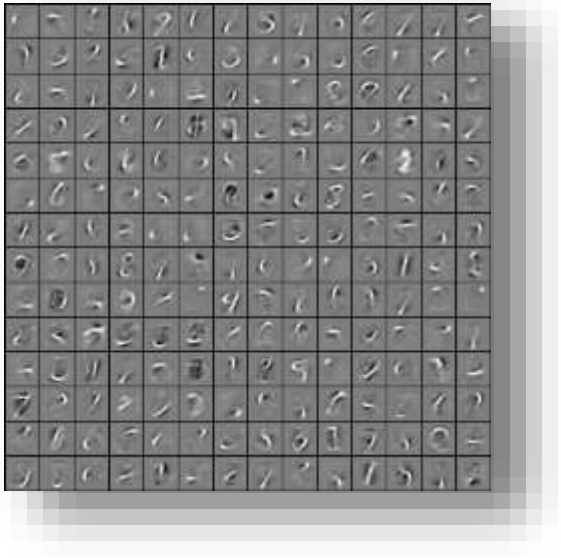
Sparse Autoencoders

Additional Reading:

Deep Learning Tutorial - Sparse Autoencoder

By Chris McCormick (2014)



Link:

http://mccormickml.com/2014/05/30/deep-learning-tutorial-sparse-autoencoder/

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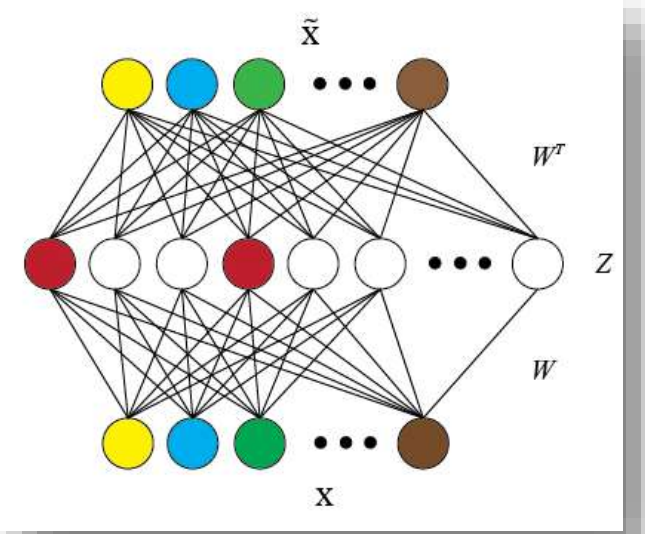
Sparse Autoencoders

Additional Reading:

Deep Learning: Sparse Autoencoders

By Eric Wilkinson (2014)



Link:

http://www.ericlwilkinson.com/blog/2014/11/19/deep-learning-sparse-autoencoders

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Sparse Autoencoders

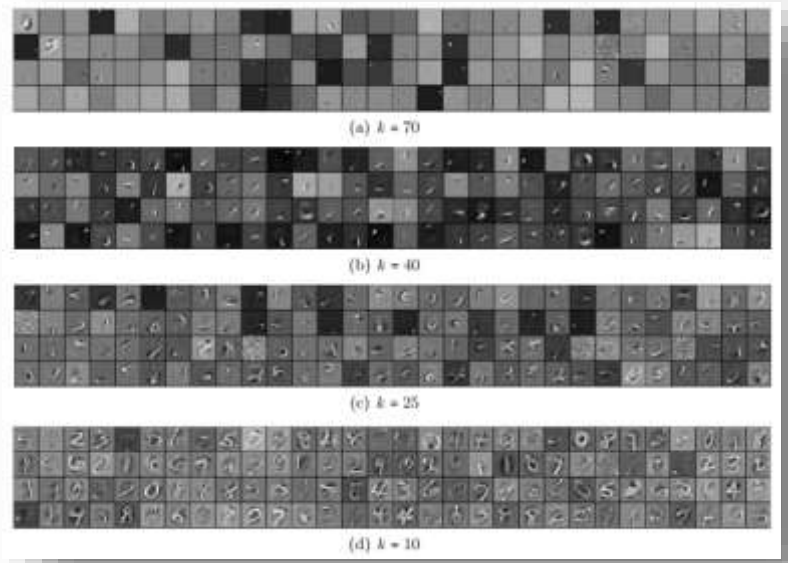
Additional Reading:

k-Sparse Autoencoders

By Alireza Makhzani et al. (2014)

Link:

https://arxiv.org/pdf/1312.5663.pdf



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Denoising Autoencoders

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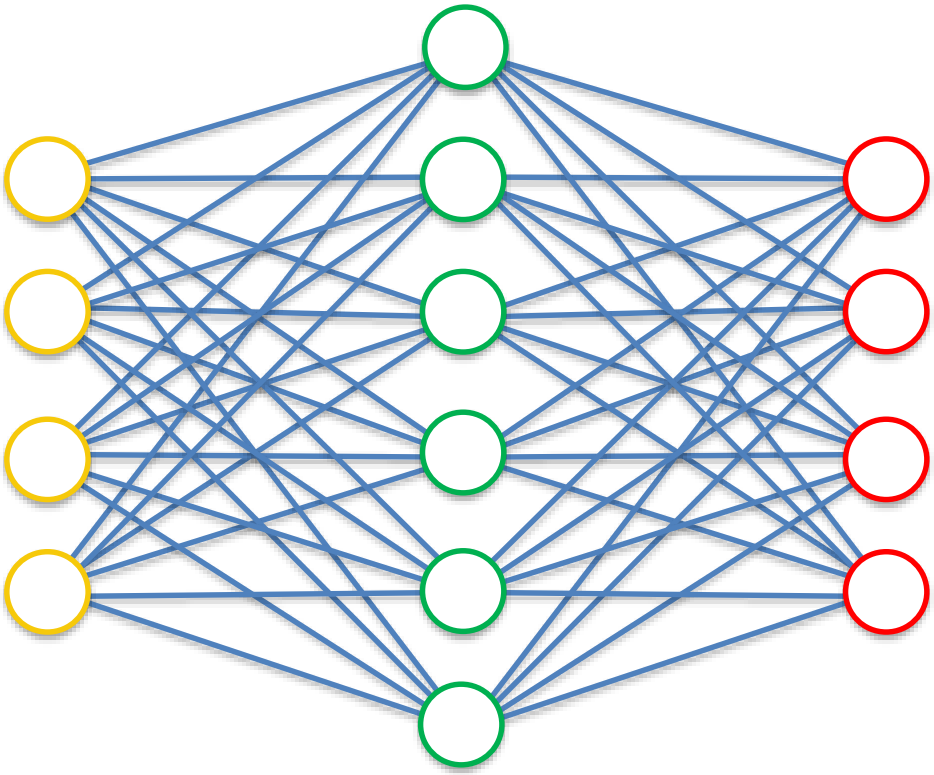
Denoising Autoencoders



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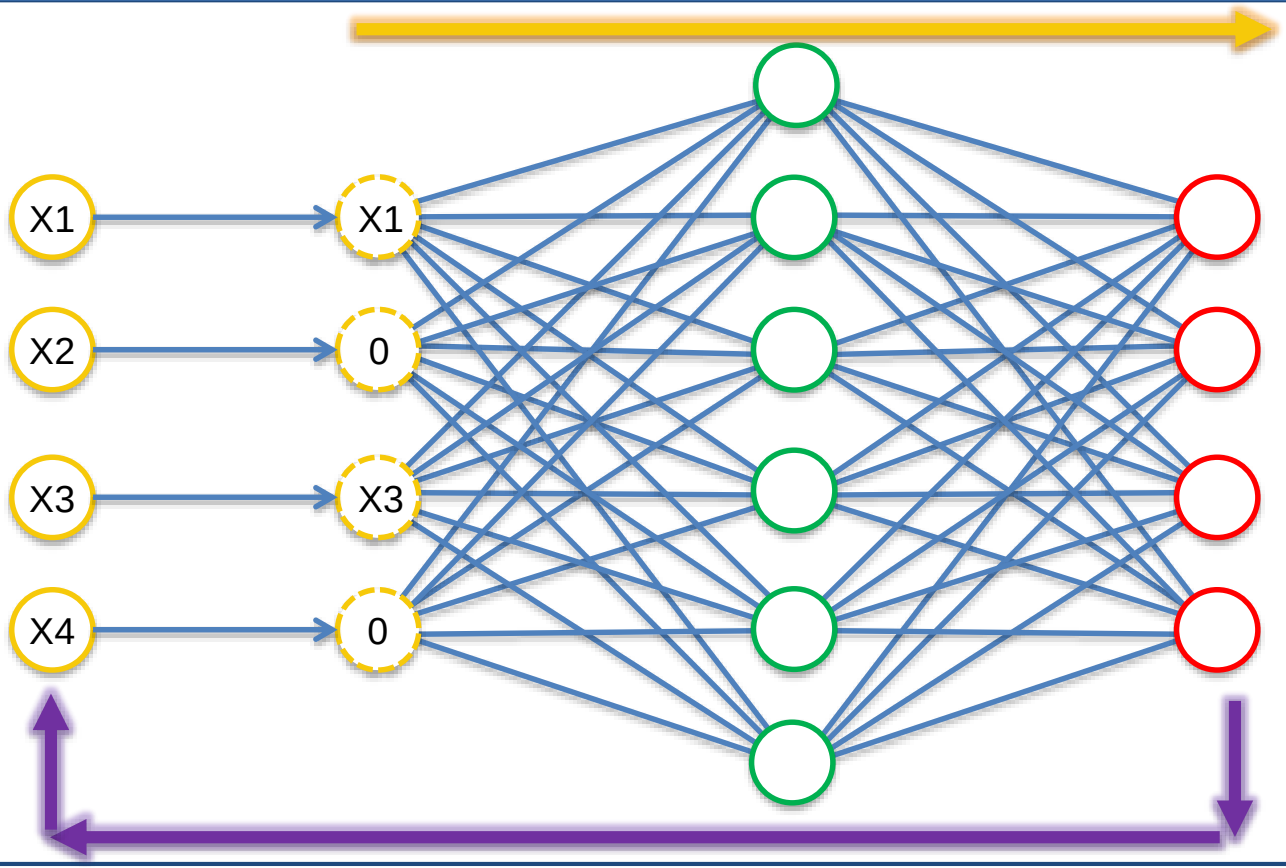
Denoising Autoencoders



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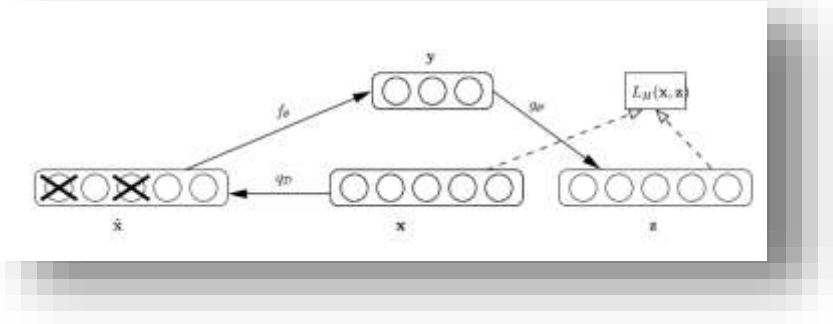
Denoising Autoencoders

Additional Reading:

Extracting and Composing Robust Features with Denoising Autoencoders

By Pascal Vincent et al. (2008)



Link:

http://www.cs.toronto.edu/~larocheh/publications/icml-2008-denoising-autoencoders.pdf

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Contractive Autoencoders

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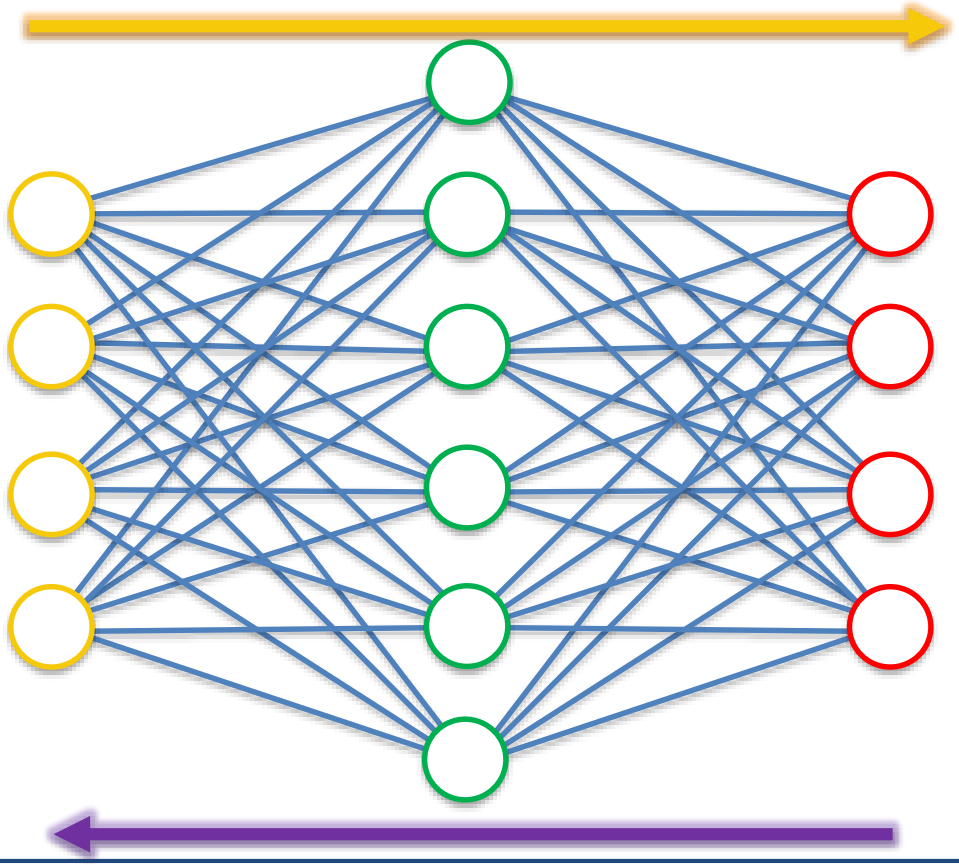
Contractive Autoencoders



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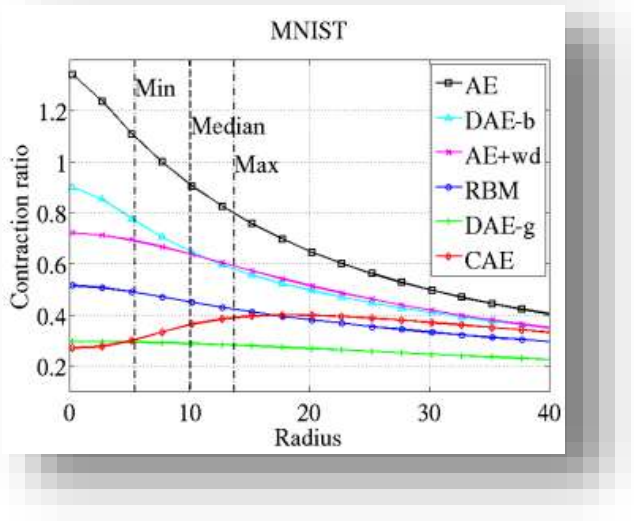
Contractive Autoencoders

Additional Reading:

Contractive Auto-Encoders: Explicit Invariance During Feature Extraction

By Salah Rifai et al. (2011)



Link:

http://machinelearning.wustl.edu/mlpapers/paper_files/ICML2011Rifai_455.pdf

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Stacked Autoencoders

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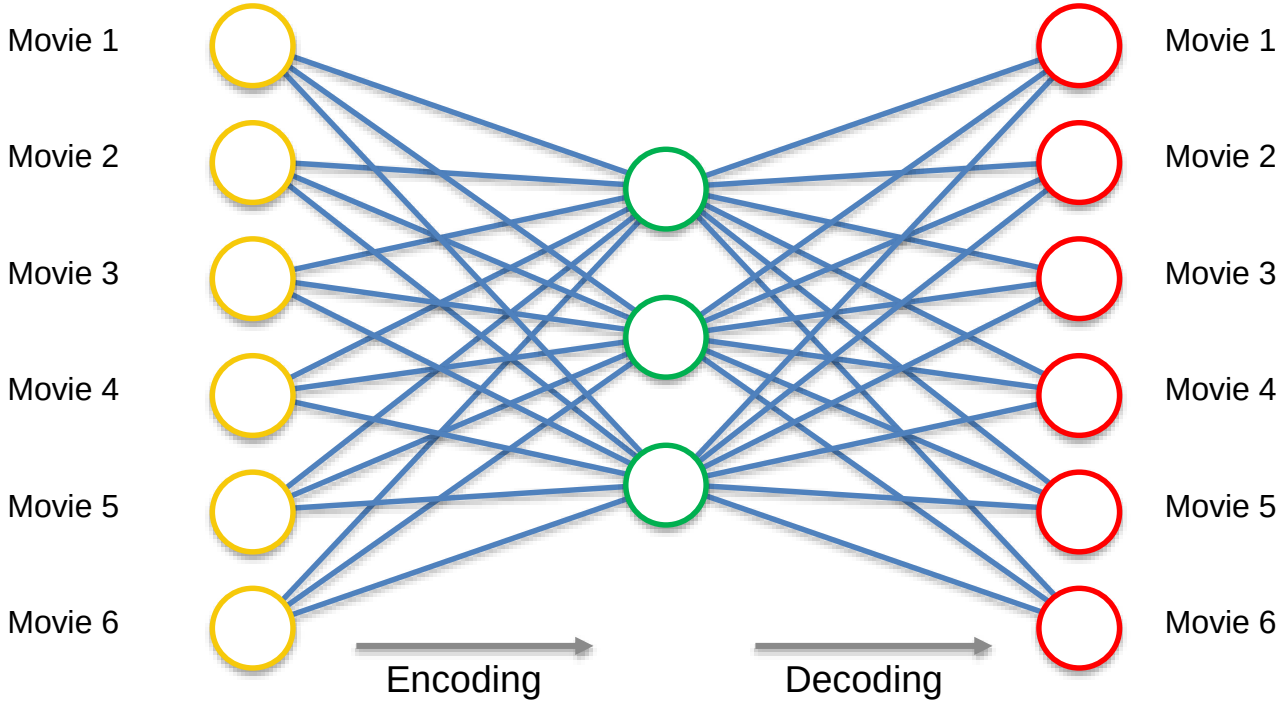
Stacked Autoencoders



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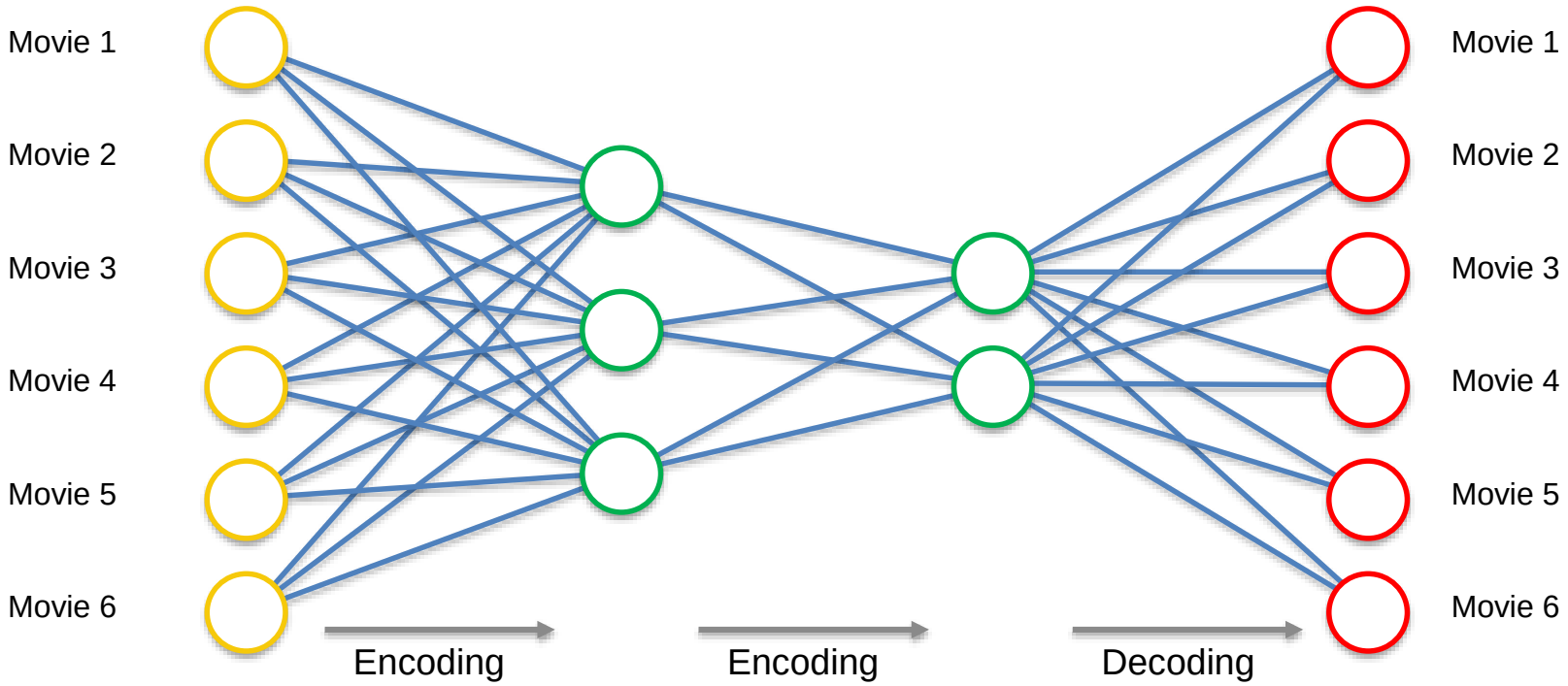
Stacked Autoencoders



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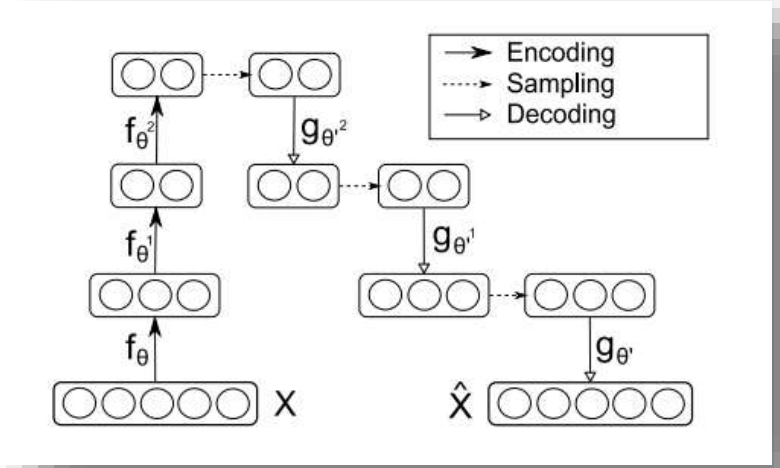
Additional Reading:

Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion

By Pascal Vincent et al. (2010)

Link:

http://www.jmlr.org/papers/volume11/vincent10a/vincent10a.pdf



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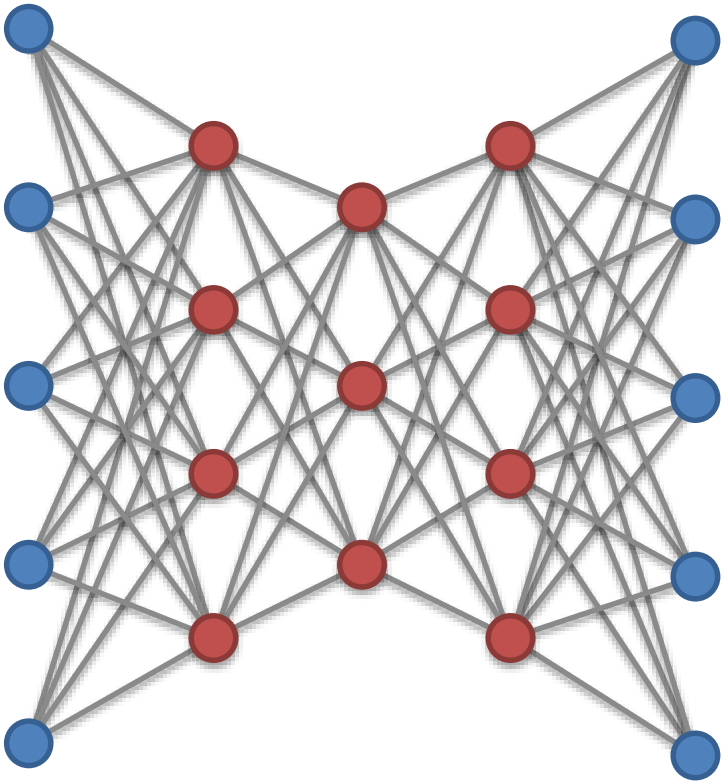

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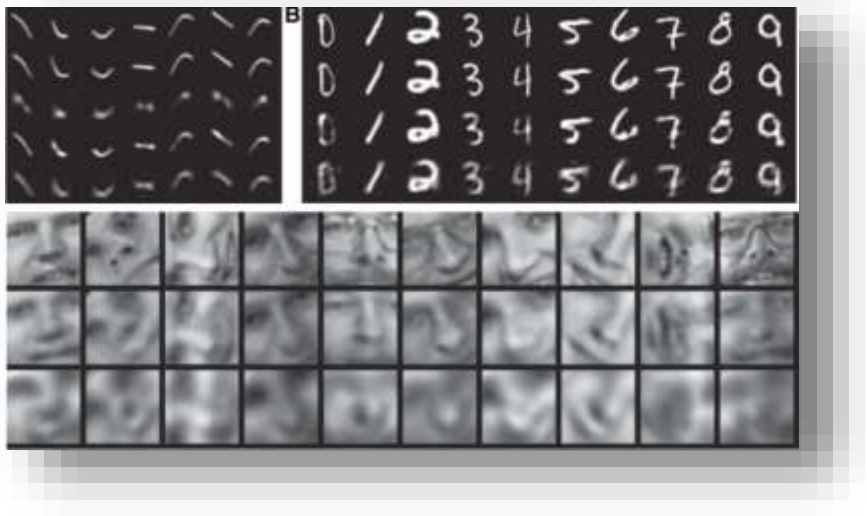
Additional Reading:

Reducing the Dimensionality of Data with Neural Networks

By Geoffrey Hinton et al. (2006)



Link:

https://www.cs.toronto.edu/~hinton/science.pdf

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